

## STRATUS TRANSACTION PROCESSING BENCHMARKS

### Introduction

There is a tendency in the computer industry today to move away from general benchmarks and performance measurements that often in their simplicity do not provide a true comparison between the combined hardware and software potential of different systems for their intended purposes.

Measures such as MIPS are often useful in comparing the CPU powers of similar machines in a single vendor's product line but seldom are useful in comparing dissimilar machines from different vendors. With multiprocessor architectures, of which the Stratus/32 is clearly an example, it is difficult to calculate the MIPS rating because MIPS will actually change as you add additional I/O processors, such as line adapters or storage directors.

Benchmarks such as the Whetstone benchmark are designed to test only a limited part of an architecture, are usually not structured for a multiprocessor architecture, and do not measure disk I/O and communications, two important parts of a transaction processing environment.

The scenario is further complicated by the reasonable requirement that with a fault tolerant architecture, the test must be run both in a "fully operational" state and a "degraded" state. Due to the hardware-based fault tolerant design of the Stratus/32, these two states are the same.

### Transaction-Based Benchmarks

Just as the Whetstone benchmark was designed to measure the scientific processing capabilities of a single CPU processor, new benchmarks were developed to test the transaction throughput of online transaction processing (OLTP) oriented machines. These benchmarks generally emulated one specific type of transaction processing environment and included both CPU and I/O.

In the last few years, two OLTP benchmarks have emerged through different channels. These are now commonly known as TPl and ETl. Although both benchmarks are transaction processing benchmarks, here the similarity stops.

ETl and TP1 are not the same benchmark or even a variation on the same benchmark. Each measures OLTP performance in a different way. Not understanding the differences between the two benchmarks has led to false comparisons. Below is a comparison between the two benchmarks:

| Operation                         | ETl  | TP1    |
|-----------------------------------|------|--------|
| CPU intensive loop                | 0 ms | 150 ms |
| Read from a large indexed file    | 1    | 5      |
| Rewrite to the large indexed file | 1    | 2      |
| Read from a small relative file   | 2    | 0      |
| Rewrite to a small relative file  | 2    | 0      |
| Write of a sequential log file    | 1    | 1      |

Although the logical I/O ratio of ETl to TP1 is 7 to 8, the physical I/O ratio is far more significant. This is because all of the TP-1 I/O's are done to large, indexed files, while most of the ETl I/O's are done to small relative files. In most modern operating systems like Stratus VOS, the disk cache will be able to eliminate physical I/O's to a small, relative file far more easily than to a large, indexed file. The "largeness" of the indexed file will guarantee at least two physical I/Os per read while the "smallness" of the relative file will almost guarantee no physical I/O's per read. Since both benchmarks do one sequential log write, this can be safely eliminated from comparison. TP1 does 12 physical I/O's per transaction to 3 in ET-1. Therefore, TP1 does four times as many physical I/O's as ETl.

### ETl

The ETl benchmark simulates a banking deposit/withdrawal transaction. It is described in an article by "anon et al" in the April 1, 1985 issue of Datamation titled "A Measure of Transaction Processing Performance".

The ETl benchmark on a single Stratus module consisted of the following:

- Read one 100 byte 3270 message from an X.25 comm line
- One indexed read from a 2,000,000 record account database
- One rewrite of that account record
- One read and rewrite of a 2,000 record teller database
- One read and rewrite of a 200 record branch database
- One write append to a sequential log file
- Format a 3270 buffer
- Send one 200 byte X25/3270 message

Written in PL1 using a requester/server model with Stratus TPF.

The Datamation article states that in 85% of the cases, the branch portion of the account number will match the current process branch number, and in the remaining 15%, the account number will reference a different process' branch. In the multimodule case, these remaining 15% were dispatched equally to different account files on modules in the system. This stimulates the model that any customer can arrive at any teller in any branch, but will go to that customer's local branch 85% of the time.

### TP1

The TP1 benchmark was designed by Stratus based on a specific international customer's requirements. It has been documented in Stratus publications and in the press. Here is a summary:

A 150ms CPU loop simulating screen form handling  
Five indexed reads from a 100,000 record database  
Rewrites of two of those records  
One write append to a sequential log file

Written in PL1 (requester) and COBOL (server) using Stratus TPF.

### Configurations:

Per XA 600:

XA 600 processor  
8 MB logical memory  
4 448 MB disks (ET-1)  
6 448 MB disks (TP-1)

### Stratus Benchmark Results

The results of TP1 and ET1 are shown in the table below. In each test, the Stratus computer was run with one of each component type removed to simulate a failure condition. It is important to understand that these results represent performance under a catastrophic hardware failure scenario such as a power supply failure, resulting in one failure per pair of StrataLINK, simplex disks, CPU, memory, and I/O controllers.

### ETl Results

| CPU Type | Module<br>Quantity | Database Size<br>(millions of<br>records) | Trans/<br>Second | 95%<br>Response Time<br>(seconds) |
|----------|--------------------|-------------------------------------------|------------------|-----------------------------------|
| XA600    | 1                  | 2M                                        | 15.2             | 0.31                              |
| XA600    | 2                  | 4M                                        | 30.0             | 0.33                              |
| XA600    | 3                  | 6M                                        | 45.0             | 0.30                              |

#### Notes:

Database size refers to the number of account records. Teller and branch records grow proportional to this.

The database is spread equally across the modules.

### TP1 Results

| CPU Type | Quantity | Transactions/Second | Response Time<br>(seconds) |
|----------|----------|---------------------|----------------------------|
| XA600    | 1        | 6.1                 | 1.9                        |

In both the ETl and TP1 cases, a prerelease version of VOS 5.1 was used.

### Conclusion

ET-1 and TP-1 are two measures of transaction processing performance. ET-1 results in significantly more transactions per second because it has fewer I/O's to accomplish. The linear growth of the Stratus architecture is clearly demonstrated by the ET-1 results, while showing excellent response times.